

Valuation Models in Disguise

Research Design Module 2

Easton and Zmijewski (1989), *Journal of Accounting and Economics*

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Name the valuation model

Two regressions the literature runs every day:

$$P_{jt} = a + b E_{jt} + \varepsilon_{jt} \quad \text{price on earnings}$$

$$\frac{P_{jt} - P_{jt-1}}{P_{jt-1}} = a + \text{ERC} \cdot \frac{E_{jt} - \mathbb{E}[E_{jt}]}{P_{jt-1}} + \varepsilon_{jt} \quad \text{returns on unexpected earnings}$$

Each one is a valuation model in disguise.

Which model, and in what kind of world does it hold?

EZ open with the general model: no accounting variable is privileged

Easton and Zmijewski's eq. (1), from Garman and Ohlson (1980):

$$P_{jt} = C_{j0} + C_{j1}A_{jt} + C_{j2}Z_{2jt} + \cdots + C_{j\Psi}Z_{\Psi jt}$$

“These models demonstrate that (stock) price is a function of all information variables that predict dividends.”

EZ (1989, p. 119), on Garman and Ohlson (1980); emphasis added

Under uncertainty, earnings (A_{jt}) is one information variable among Ψ . Every response coefficient depends on risk and on the entire system of time-series processes.

The pivot to capitalization happens in a footnote

“Assuming the general system of time-series processes in Garman and Ohlson (1980) the signs of the partial derivatives cannot even be evaluated because the response coefficients (ERC) are too complex. Under some restrictive assumptions, however, the signs... can be evaluated ... One set of assumptions that yield these [sign] predictions are the assumptions⁸ underlying the earnings capitalization model [Miller and Modigliani (1961)].”

EZ (1989, pp. 120–121)

Footnote 8, in full: *“See Ohlson (1983) for a discussion of these assumptions.”*

The capitalization equation is never written down: no $P = E/r$ appears anywhere in the paper.

Both workhorse models describe a world of certainty

Earnings capitalization

$$MV = E/r$$

Requires: future payoffs known; earnings permanent (no transitory component, no growth); investment earns exactly r (Miller and Modigliani 1961).

Book value capitalization

$$MV = BV$$

Requires: accounting records assets at value (unbiased, mark-to-market book), so book earns exactly r and expected residual income is zero.

Under uncertainty, each holds only in a knife-edge case.

The footnoted paper says the case is knife-edge

Ohlson (1983), “Price-Earnings Ratios and Earnings Capitalization Under Uncertainty” (*JAR*), built directly on Garman and Ohlson (1980) and Rubinstein (1976):

- Price equals capitalized earnings under uncertainty *only if* earnings are a sufficient statistic for the payoff process (random-walk-with-payout dynamics)
- A constant price–earnings ratio obtains only in that case

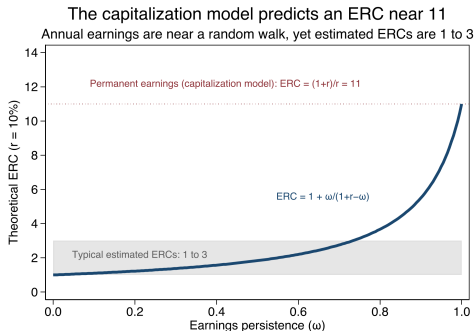
A 53-citation paper carries the functional form of the ERC literature.

The regression you run is the valuation model you assume

Regression	Implied model	World required
Price on earnings (levels)	$MV = E/r$	certainty, permanent earnings
Price on book value	$MV = BV$	certainty, unbiased accounting
Returns on unexpected earnings (ERC)	$\Delta \text{ of } MV = E/r$	same, in first differences
Returns on levels and changes (Easton-Harris 1991)	both polar models	both certainty worlds, mixed

Kothari and Zimmerman (1995) say it explicitly: their price model *is* capitalized random-walk earnings, and the return model is its first difference.

Take the model seriously and it misses by a factor of four to seven



EZ's own estimates: 1.65 and 2.53. Their concession: “the range of forms (and, therefore, magnitudes) of the ERC is infinite” (p. 121).

Under uncertainty, the certainty models survive only as polar cases

Garman and Ohlson (1980): arbitrage-free valuation under uncertainty

Clean surplus: $MV = BV + PV(\text{residual income})$ (Peasnell 1982)

Ohlson (1995): $P = k(\varphi E - d) + (1 - k) BV$

$k = k(\omega)$ rises with persistence; $\varphi = (1 + r)/r$

$$MV = E/r \quad (\omega \rightarrow 1)$$

$$MV = BV \quad (\omega \rightarrow 0)$$

Price becomes a weighted average of capitalized earnings and book value; the certainty models are its endpoints.

A lens that focuses only under certainty cannot read magnitudes

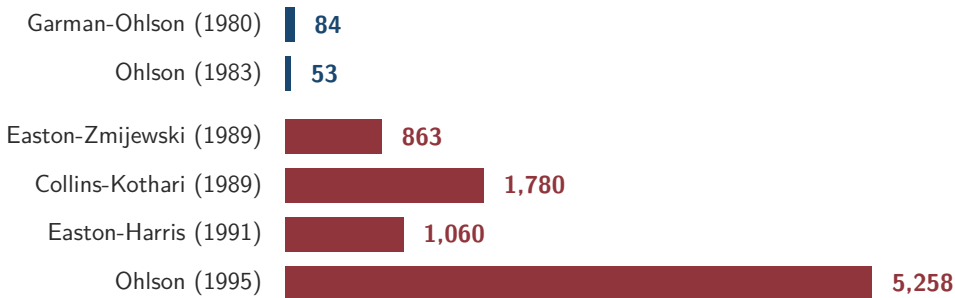
The defense: valuation models are organizing lenses, not literal truth (paraphrasing Barth, Beaver, and Landsman 2001).

Fair, for signs and rankings. The moment a coefficient is read as a magnitude (is an ERC of 2 small?), the lens must hold in the world that generated the data (Holthausen and Watts 2001).

EZ themselves claim only signs.

Their hypotheses are comparative statics (ERC increasing in persistence, decreasing in risk), and they warn the ERC's form is not pinned down. The magnitude interpretations came later, and inherited the footnote.

The general theory was cited, then quietly abandoned



Citations (OpenAlex, August 2026); navy = the papers carrying the assumptions

Kothari-Zimmerman (1995), Holthausen-Watts (2001), and Barth-Beaver-Landsman (2001) cite Garman-Ohlson zero times.

Summary

- Price-on-earnings and price-on-book-value regressions embed the earnings-capitalization and book-value models; both obtain under certainty, and only in knife-edge cases under uncertainty (Ohlson 1983).
- EZ motivate their predictions with the general Garman-Ohlson model but delegate the capitalization assumptions to a footnote; the model doing the empirical work is never stated.
- Sign predictions survive these restrictions; magnitude interpretations do not. A value-relevance specification should state its valuation model and evaluate its assumptions rather than delegate them to citation.

Garman and Ohlson (1980) · Ohlson (1983, 1995) ·
github.com/sbb151/acctg597a-research-design